

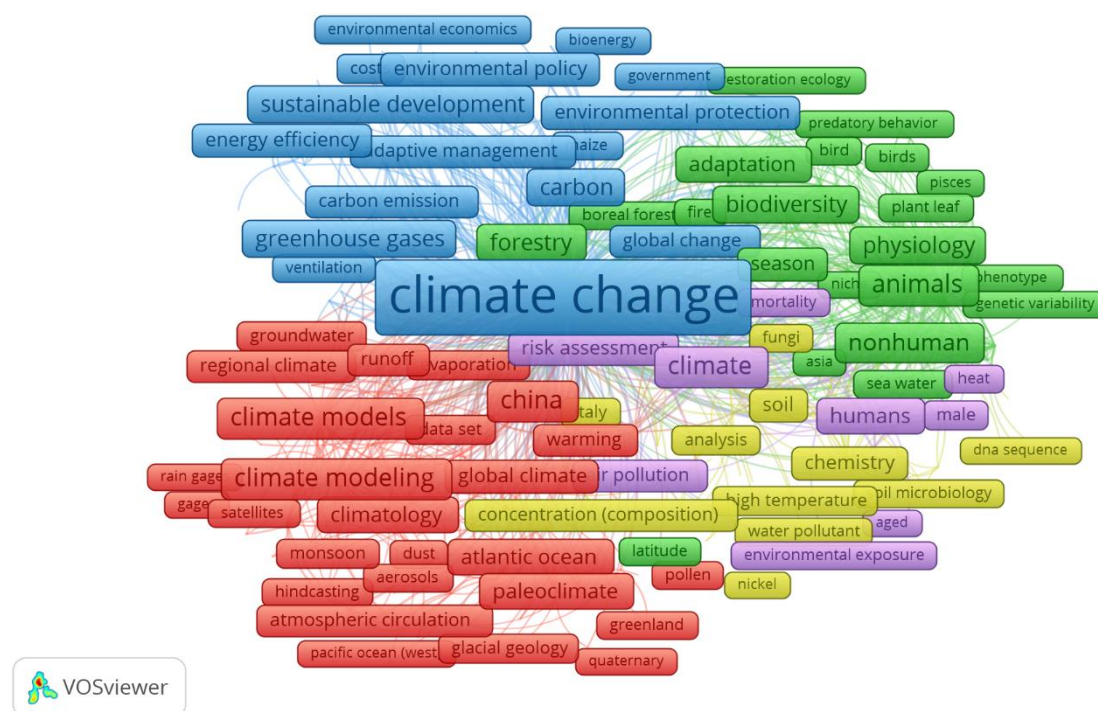
SDG13: Take urgent action to combat climate change and its impacts

SUMMARY

Original search string: 161,362 document results

TITLE-ABS-KEY(((climate action} OR {climate adaptation} OR {climate change} OR IPCC OR {climate effect} OR {climate feedback} OR {climate forcing} OR {climate impact} OR {climate mitigation} OR {climate model} OR {climate models} OR {climate modeling} OR {climate modelling} OR {climate policy} OR {climate policies} OR {climate risk} OR {climate risks} OR {climate services} OR {climate service} OR {climate prediction} OR {climate predictions} OR {climate signal} OR {climate signals} OR {climate tipping point} OR {climate variation} OR {climate variations} OR ecoclimatology OR eco-climatology OR {regional climate} OR {regional climates} OR {urban climate} OR {urban climates} OR (climate AND (adaptation OR {adaptive management} OR biodivers* OR bioeconomy OR {biological production} OR carbon OR conservation OR emission* OR extreme OR {food chain} OR {food chains} OR hazard* OR island* OR {land use} OR megacit* OR mitigation OR consumption OR production) OR anthropocene OR atmospher* OR {clean development mechanism} OR {glacier retreat} OR warming OR greenhouse OR {ice-ocean interaction} OR {ice-ocean interactions} OR {nitrogen cycle} OR {nitrogen cycles} OR {ocean acidification} OR {radiative forcing} OR {sea ice} OR {sea level} OR {sea levels} OR {thermal expansion} OR UNFCCC OR ozone))) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



Updated search query (v2): 157,241 document results

TITLE-ABS-KEY (({climate action} OR {climate adaptation} OR {climate change} OR {climate capitalism} OR ipcc OR {climate effect} OR {climate equity} OR {climate feedback} OR {climate finance} OR {climate change financing} OR {climate forcing} OR {climate governance} OR {climate impact} OR {climate investment} OR {climate justice} OR {climate mitigation} OR {climate model} OR {climate models} OR {climate modeling} OR {climate modelling} OR {climate policy} OR {climate policies} OR {climate risk} OR {climate risks} OR {climate services} OR {climate service} OR {climate prediction} OR {climate predictions} OR {climate signal} OR {climate signals} OR {climate tipping point} OR {climate variation} OR {climate variations} OR ecoclimatology OR eco-climatology OR {Green Climate Fund} OR {regional climate} OR {regional climates} OR {urban climate} OR {urban climates} OR (climate AND ({adaptive management} OR awareness OR bioeconomy OR carbon OR {decision-making} OR {disaster risk reduction} OR {environmental education} OR {sustainable development education} OR {energy conservation} OR emission* OR extreme OR {food chain} OR {food chains} OR framework OR hazard* OR island* OR {land use} OR megacit* OR consumption OR production OR {small island developing states} OR anthropocene OR atmospher* OR {clean development mechanism} OR {glacier retreat} OR warming OR greenhouse OR {ice-ocean interaction} OR {ice-ocean interactions} OR {nitrogen cycle} OR {nitrogen cycles} OR {ocean acidification} OR {radiative forcing} OR {sea ice} OR {sea level} OR {sea levels} OR {thermal expansion} OR unfccc OR ozone))) AND NOT ({drug} OR {geomorphology})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

PROCESS

The first stage of this review was to compare the original search string to the targets listed under the Sustainable Development Goal (SDG).

SDG targets listed here: <https://sustainabledevelopment.un.org/sdg13>

The purpose of this stage was to understand whether the Scopus results correctly reflected the theme of the SDG. The SDGs were unanimously agreed by member states of the United Nations, with very deliberate language and scope/ exclusions, so the results of this analysis should accurately reflect the specific SDG themes.

SDG 13 promotes the integration of **climate change adaptation and mitigation** measures into (inter)national policies, strategies and planning to **strengthen resilience** and **adaptive capacity** to hazards and natural disasters induced by climate change. It promotes **awareness raising and education** as well as institutional and **individual capacity building** regarding climate change mitigation, adaptation and impact reduction.

A Scopus search was conducted using the original search string. This returned 161,362 document results.

The top 20 cited articles were reviewed. Five of the 20 articles do not closely reflect the theme of the SDG. Specifically, these articles addressed issues regarding land use, land use change, ecology or medicine.

Top 20 most cited articles listed in Scopus from the original search string:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3500	2-s2.0-84926664150
High-resolution global maps of 21st-century forest cover change	2013	Science	2830	2-s2.0-84887724460
Updated high-resolution grids of monthly climatic observations - the CRU TS3.10 Dataset	2014	International Journal of Climatology	2469	2-s2.0-84896707401
Bounding the role of black carbon in the climate system: A scientific assessment	2013	Journal of Geophysical Research Atmospheres	1995	2-s2.0-84875359259
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1624	2-s2.0-84922433390
Köppen's climate classification map for Brazil	2013	Meteorologische Zeitschrift	1528	2-s2.0-84897100402
High secondary aerosol contribution to particulate pollution during haze events in China	2015	Nature	1422	2-s2.0-84908317345

Karst Hydrogeology and Geomorphology	2013	Karst Hydrogeology and Geomorphology	1343	2-s2.0-84949745121
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1274	2-s2.0-84871840877
Anthropogenic and natural radiative forcing	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1265	2-s2.0-84928644022
Changes in the global value of ecosystem services	2014	Global Environmental Change	1237	2-s2.0-84901406973
Antibiotic resistance-the need for global solutions	2013	The Lancet Infectious Diseases	1210	2-s2.0-84887627398
Long-term climate change: Projections, commitments and irreversibility	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1199	2-s2.0-84928600768
Porous materials with optimal adsorption thermodynamics and kinetics for co2 separation	2013	Nature	1126	2-s2.0-84874745873
Photocatalytic reduction of CO2 on TiO2 and other semiconductors	2013	Angewandte Chemie - International Edition	1072	2-s2.0-84880131318
The JRA-55 reanalysis: General specifications and basic characteristics	2015	Journal of the Meteorological Society of Japan	954	2-s2.0-84920667419
Observations: Atmosphere and surface	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	924	2-s2.0-84928600517
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental	915	2-s2.0-84928642948

		Panel on Climate Change		
Status and ecological effects of the world's largest carnivores	2014	Science	911	2-s2.0-84892185014
The NCEP climate forecast system version 2	2014	Journal of Climate	900	2-s2.0-84896093136

In addition, a randomised sample of 20 articles were reviewed. Two of the 20 articles do not closely reflect the theme of the SDG. These two articles related to socio-economic improvements for farmers / foresters and earth sciences on erosion and weathering.

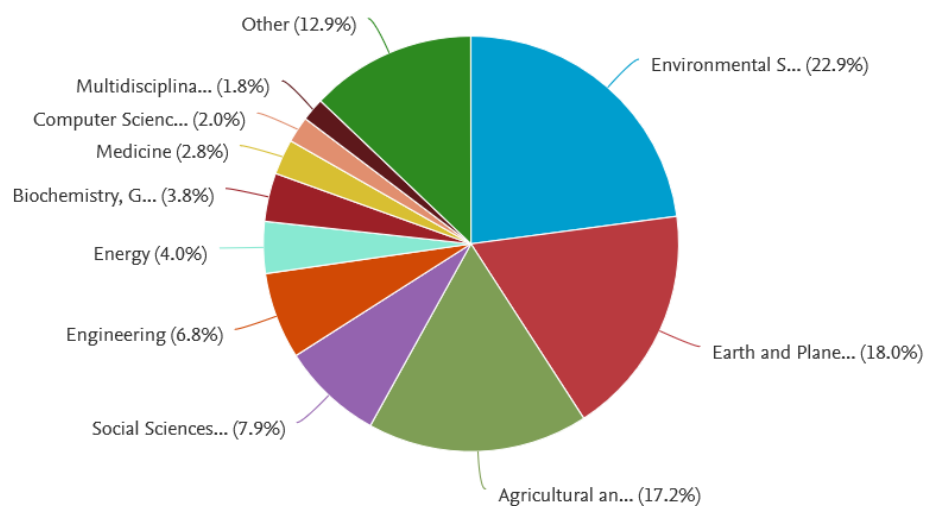
Randomised sample of 20 articles listed in Scopus from the original search string:

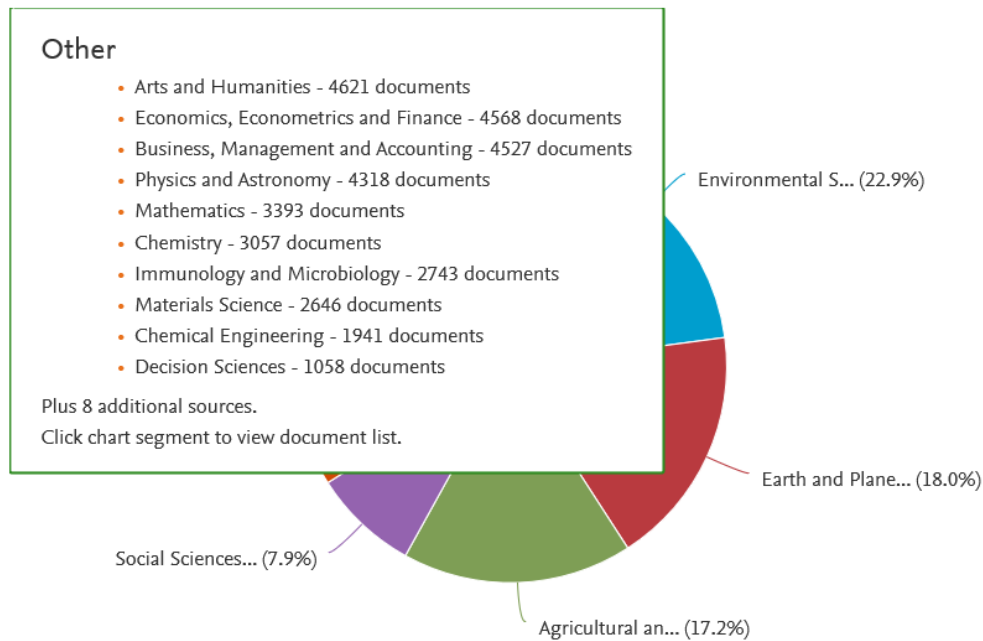
*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Permafrost collapse alters soil carbon stocks, respiration, CH ₄ , and N ₂ O in upland tundra	2015	Global Change Biology	44	2-s2.0-84946780354
Assessing the CAM5 physics suite in the WRF-Chem model: Implementation, resolution sensitivity, and a first evaluation for a regional case study	2014	Geoscientific Model Development	42	2-s2.0-84900454585
Stability of Soil Carbon Stocks Varies with Forest Composition in the Canadian Boreal Biome	2013	Ecosystems	39	2-s2.0-84880293056
Optimizing Deficit Irrigation Scheduling Under Shallow Groundwater Conditions in Lower Reaches of Amu Darya River Basin	2013	Water Resources Management	26	2-s2.0-84876988410
WORLD CLIMATE: A Role-Play Simulation of Climate Negotiations	2015	Simulation and Gaming	24	2-s2.0-84947264046
SEA making inroads in land-use planning in Brazil: The case of the Extreme South of Bahia with forestry and biofuels	2013	Land Use Policy	17	2-s2.0-84880993212
The Asian monsoon and its future change in climate models: A review	2017	Journal of the Meteorological Society of Japan	12	2-s2.0-85014085970
Food stocks and grain reserves: evaluating whether storing food creates resilient food systems	2015	Journal of Environmental Studies and Sciences	10	2-s2.0-84977083768
SPOT-VEGETATION GEOV1 biophysical parameters in semi-arid agro-ecosystems	2014	International Journal of Remote Sensing	8	2-s2.0-84897057132
Effects of climate change on household economy and adaptive responses among agricultural households in eastern coast of India	2016	Current Science	5	2-s2.0-84969622671
Water and People: Assessing Policy Priorities for Climate Change Adaptation in the Mediterranean	2013	Advances in Global Change Research	5	2-s2.0-85015719621
Chemical and physical weathering in a hot-arid, tectonically active alluvial system of anza borrego desert, california	2016	Sedimentology	4	2-s2.0-84959136820

Spatiotemporal properties of growing season indices during 1961–2010 and possible association with agroclimatological regionalization of dominant crops in Xinjiang, China	2016	Meteorology and Atmospheric Physics	4	2-s2.0-84948651215
Seasonal drought predictability in Portugal using statistical–dynamical techniques	2016	Physics and Chemistry of the Earth	4	2-s2.0-84929791781
The interplay between risk attitudes and low probability, high cost outcomes in climate policy analysis	2013	Environmental Modelling and Software	4	2-s2.0-84872025493
Developing the prototype AndaLAND for agriculture soil and water assessment in climate change scenarios [Desenvolvimento do protótipo AndaLAND para avaliação de solos agrícolas e da água em cenários de alterações climáticas] [Desarrollo del prototipo AndaLAND para evaluar suelos agrícolas y agua en el marco del cambio climático]	2016	Spanish Journal of Soil Science	1	2-s2.0-84962255399
Global air-sea surface carbon dioxide transfer velocity and flux estimated using 17 a altimeter data and a new algorithm	2013	Acta Oceanologica Sinica	1	2-s2.0-84890163130
What does climate change adaptation mean for humanitarian assistance? Guiding principles for policymakers and practitioners	2017	IDS Bulletin	0	2-s2.0-85031796934
Conservation Agriculture Among Farmers in Kalomo, Zambia: Potential for Productivity Under Climate Change	2017	Beyond Agricultural Impacts: Multiple Perspectives on Climate Change and Agriculture in Africa	0	2-s2.0-85054284613
Predicting impacts of climate change on Chinese hybrid poplar using maxent modeling	2014	Applied Mechanics and Materials	0	2-s2.0-84914689081

Documents by subject area





The field of climate change is a broad and multidisciplinary field. Articles on climate change and climate change adaptation/mitigation are diverse and can comprise a large variety of topics, ranging from social sciences on policies and behavioural studies to environmental or agricultural studies on land uses, food system adaptation and climate forecasts. This is reflected in general analysis of the document results and the resulting pie charts, with a large number of articles retrieved that are classified in the field of 'Environmental Sciences', 'Earth and Planetary Sciences', 'Agricultural and Biological Sciences', 'Social Sciences' and 'Engineering'. Articles classified in these fields can all be relevant for the theme of the SDG. The original search query returns a low number of irrelevant articles published in the field of land use, earth sciences and medicine. These could be excluded by narrowing down keywords and excluding specific fields (e.g. the field of medicine) in future updates.

Update 1) The search keywords were edited to remove the broader search terms, or to make them more specific by combining with other search keywords.

Original search string: 161,362 document results

TITLE-ABS-KEY(((climate action} OR {climate adaptation} OR {climate change} OR IPCC OR {climate effect} OR {climate feedback} OR {climate forcing} OR {climate impact} OR {climate mitigation} OR {climate model} OR {climate models} OR {climate modeling} OR {climate modelling} OR {climate policy} OR {climate policies} OR {climate risk} OR {climate risks} OR {climate services} OR {climate service} OR {climate prediction} OR {climate predictions} OR {climate signal} OR {climate signals} OR {climate tipping point} OR {climate variation} OR {climate variations} OR ecoclimatology OR eco-climatology OR {regional climate} OR {regional climates} OR {urban climate} OR {urban climates} OR (climate AND (adaptation OR {adaptive management} OR biodivers* OR bioeconomy OR {biological production} OR carbon OR conservation OR emission* OR extreme OR {food chain} OR {food chains} OR hazard* OR island* OR {land use} OR megacit* OR mitigation OR consumption OR production) OR anthropocene OR atmospher* OR {clean development mechanism} OR {glacier retreat} OR warming OR greenhouse OR {ice-ocean interaction} OR {ice-ocean interactions} OR {nitrogen cycle} OR {nitrogen cycles} OR {ocean acidification} OR {radiative forcing} OR {sea ice} OR {sea level} OR {sea levels} OR {thermal expansion} OR UNFCCC OR ozone))) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

Rationale for edits to the search keywords:

query term	decision	rationale
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Conservation	Changed to {energy conservation}	Conservation as a keyword itself, even in combination with climate, can also lead to hits on biodiversity conservation / studies on ecology, which do not closely relate to the theme of this SDG. The keyword was narrowed down to {energy conservation} to ensure climate change related hits, as energy conservation is one of the main targets to adapt to new institutional and production systems to mitigate climate change effects
{Biodivers*}	Deleted	Biodiversity is not closely related to the theme of this SDG. Whereas biodiversity is affected by climate change, this theme covers climate change itself, as well as societal mitigation and adaptation but not necessarily natural resources and biodiversity (except for mitigation in the field of agriculture, forestry, etc). Excluded to reduce hits on studies on biodiversity and ecology
AND NOT {drug}	ADDED	AND NOT {drug}, to reduce hits on articles in the field of medicine.
AND NOT {geomorphology}	ADDED	AND NOT {geomorphology}, to reduce hits on earth sciences / sedimentary sciences that study changes in earth layers / sediments, partly to identify historic climate change effects, but these are not directly related to the current societal challenge of adapting to climate change
awareness	ADDED to (climate AND ...)	To include targets on education
education	ADDED to (climate AND ...)	To include targets on education
{decision-making}	ADDED to (climate AND ...)	To include targets on decision-making and policy-making

Updated search string (first update): 158.963 document results

TITLE-ABS-KEY(((climate action} OR {climate adaptation} OR {climate change} OR IPCC OR {climate effect} OR {climate feedback} OR {climate forcing} OR {climate impact} OR {climate mitigation} OR {climate model} OR {climate models} OR {climate modeling} OR {climate modelling} OR {climate policy} OR {climate policies} OR {climate risk} OR {climate risks} OR {climate services} OR {climate service} OR {climate prediction} OR {climate predictions} OR {climate signal} OR {climate signals} OR {climate tipping point} OR {climate variation} OR {climate variations} OR ecoclimatology OR eco-climatology OR {regional climate} OR {regional climates} OR {urban climate} OR {urban climates} OR (climate AND (adaptation OR {adaptive management} OR awareness OR bioeconomy OR {biological production} OR carbon OR {decision-making} OR education OR {energy conservation} OR emission* OR extreme OR {food chain} OR {food chains} OR hazard* OR island* OR {land use} OR megacit* OR mitigation OR consumption OR production) OR anthropocene OR atmospher* OR {clean development mechanism} OR {glacier retreat} OR warming OR greenhouse OR {ice-ocean interaction} OR {ice-ocean interactions} OR {nitrogen cycle} OR {nitrogen cycles} OR {ocean acidification} OR {radiative forcing} OR {sea ice} OR {sea level} OR {sea levels} OR {thermal expansion} OR UNFCCC OR ozone)) AND NOT ({drug} OR {geomorphology})) AND PUBYEAR BEF 2018 AND PUBYEAR AFT 2012

The results of the updated search were reviewed to check whether they aligned closely with the SDG theme.

The updated search returned 158.963 document results, compared to 161,362 document results for the original search.

The top 20 most cited articles were reviewed. Three of the 20 articles do not closely match the theme of the SDG. Specifically, these related to land cover changes, threats to ecosystems or ecosystem services unrelated to climate change. One article on oceanic acidification was considered to be relevant as acidification is a result of climate change (carbon uptake leading to acidification), which causes natural hazards such as declining fish stocks / extinction. However, the abstract of the article did not deal with these impacts nor on mitigation.

Top 20 most cited articles listed in Scopus from search string version 1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3502	2-s2.0-84926664150
High-resolution global maps of 21st-century forest cover change	2013	Science	2832	2-s2.0-84887724460
Updated high-resolution grids of monthly climatic observations - the CRU TS3.10 Dataset	2014	International Journal of Climatology	2469	2-s2.0-84896707401
Bounding the role of black carbon in the climate system: A scientific assessment	2013	Journal of Geophysical Research Atmospheres	1996	2-s2.0-84875359259
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1624	2-s2.0-84922433390
Köppen's climate classification map for Brazil	2013	Meteorologische Zeitschrift	1528	2-s2.0-84897100402
High secondary aerosol contribution to particulate pollution during haze events in China	2015	Nature	1424	2-s2.0-84908317345
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1274	2-s2.0-84871840877
Anthropogenic and natural radiative forcing	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1265	2-s2.0-84928644022
Changes in the global value of ecosystem services	2014	Global Environmental Change	1237	2-s2.0-84901406973

Long-term climate change: Projections, commitments and irreversibility	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1199	2-s2.0-84928600768
Porous materials with optimal adsorption thermodynamics and kinetics for co2 separation	2013	Nature	1126	2-s2.0-84874745873
Photocatalytic reduction of CO2 on TiO2 and other semiconductors	2013	Angewandte Chemie - International Edition	1072	2-s2.0-84880131318
The JRA-55 reanalysis: General specifications and basic characteristics	2015	Journal of the Meteorological Society of Japan	955	2-s2.0-84920667419
Observations: Atmosphere and surface	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	924	2-s2.0-84928600517
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	916	2-s2.0-84928642948
Status and ecological effects of the world's largest carnivores	2014	Science	912	2-s2.0-84892185014
The NCEP climate forecast system version 2	2014	Journal of Climate	900	2-s2.0-84896093136
Carbon capture and storage update	2014	Energy and Environmental Science	869	2-s2.0-84890526899
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	826	2-s2.0-84877016612

In addition, a randomised sample of 20 articles were reviewed. Two of the 20 articles do not closely reflect the theme of this SDG. Both articles cover studies on historical climate change. It could be argued that insights in historical events could help understand current climate change events. However, these were considered to be too far off regarding the theme on mitigation / adaptation / policy-making and current effects.

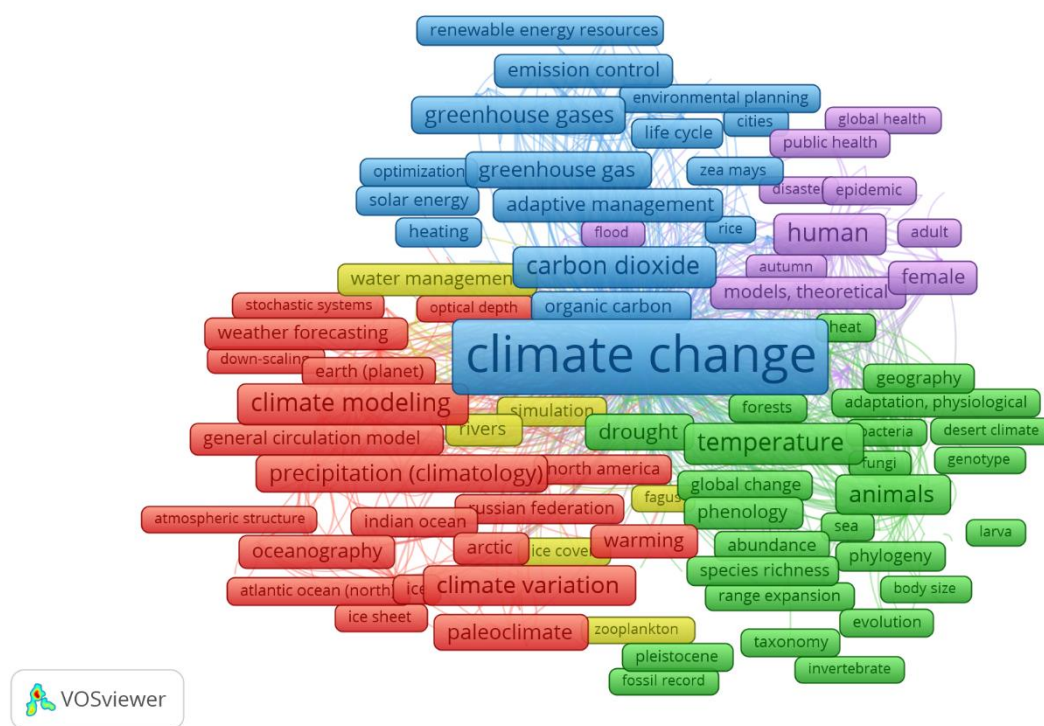
Randomised sample of 20 articles listed in Scopus from search string version 1:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Evaluating environmental impacts on tropical cyclone rapid intensification predictability utilizing statistical models	2015	Weather and Forecasting	45	2-s2.0-84944605093
Synchronous turnover of flora, fauna, and climate at the Eocene-Oligocene Boundary in Asia	2014	Scientific Reports	32	2-s2.0-84948469731
Importance of vegetation dynamics for future terrestrial carbon cycling	2015	Environmental Research Letters	23	2-s2.0-84930226901
Cloud liquid water path and radiative feedbacks over the Southern Ocean	2016	Geophysical Research Letters	19	2-s2.0-84995701641
Impact of feedstock, land use change, and soil organic carbon on energy and greenhouse gas performance of biomass cogeneration technologies	2015	Applied Energy	18	2-s2.0-84929311215
A synthesis of cloud condensation nuclei counter (CCNC) measurements within the EUCAARI network	2015	Atmospheric Chemistry and Physics	16	2-s2.0-84946238238
Climate change impacts on potential recruitment in an ecosystem engineer	2013	Ecology and Evolution	12	2-s2.0-84886236995
Mesoamerican biological corridor	2014	Climate and Conservation: Landscape and Seascape Science, Planning, and Action	8	2-s2.0-84899751143
Evaluation of daily maximum and minimum 2-m temperatures as simulated with the regional climate model COSMO-CLM over Africa	2013	Meteorologische Zeitschrift	8	2-s2.0-84887836806
Coral calcification under daily oxygen saturation and pH dynamics reveals the important role of oxygen	2014	Biology Open	7	2-s2.0-84979072974
Are temperate land snails susceptible to climate change through reduced altitudinal ranges? A pennsylvania example	2013	American Malacological Bulletin	7	2-s2.0-84882443798
Perspectives: Legislating change	2013	Nature	5	2-s2.0-84885088816
Articulating social science in the wild of global natures? on economics and anthropology in transnational environmental politics	2014	Environment and Planning A	3	2-s2.0-84907297791
Economic and Environmental Impact Trade-Offs Related to In-Water Hull Cleanings of Merchant Vessels	2017	Journal of Industrial Ecology	2	2-s2.0-85024109544
Potential impact of climate change on late blight outbreak in western Uttar Pradesh and Punjab using JHULSACAST model	2015	Potato Journal	2	2-s2.0-84948975471
The Spatial Distribution of Precipitation in Northwest China	2014	Journal of Electrical and Computer Engineering	2	2-s2.0-84937019925

Late glacial and early Holocene hydroclimate variability in northwest Iran (Talesh Mountains) inferred from chironomid and pollen analysis	2017	Journal of Paleolimnology	1	2-s2.0-85020104319
A comparative study on effects of spatial aggregation for GlobeLand30	2016	International Conference on Geoinformatics	1	2-s2.0-84962367708
Exploring the water-food-energy and climate nexus: Insights from the moroccan draa valley	2017	Springer Tracts in Civil Engineering	0	2-s2.0-85060285006
Arctic Alaska, USA	2014	Climate and Conservation: Landscape and Seascape Science, Planning, and Action	0	2-s2.0-84937897361

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the original search results to check if the updated search was primarily removing irrelevant results.

The original and updated searches were combined using AND NOT function in Scopus. This returned 5,399 document results – i.e. the publications that were excluded from the first search in the second search.

The top 20 most cited articles were reviewed. Three of the 20 articles do not closely reflect the theme of the SDG. Seventeen of the 20 articles do not closely reflect the theme of the SDG. Three articles are excluded from the updated search results which are relevant for the SDG.

Title	Year	Source title	Cited by	EID
Karst Hydrogeology and Geomorphology	2013	Karst Hydrogeology and Geomorphology	1343	2-s2.0-84949745121
Antibiotic resistance-the need for global solutions	2013	The Lancet Infectious Diseases	1210	2-s2.0-84887627398
Bee declines driven by combined Stress from parasites, pesticides, and lack of flowers	2015	Science	690	2-s2.0-84927566064
Eukaryotic plankton diversity in the sunlit ocean	2015	Science	471	2-s2.0-84929996994
Tidal wetland stability in the face of human impacts and sea-level rise	2013	Nature	446	2-s2.0-84889592676
Large contribution of natural aerosols to uncertainty in indirect forcing	2013	Nature	348	2-s2.0-84887332636
Mapping species distributions with MAXENT using a geographically biased sample of presence data: A performance assessment of methods for correcting sampling bias	2014	PLoS ONE	289	2-s2.0-84901308299
Complete biosynthesis of opioids in yeast	2015	Science	284	2-s2.0-84941346066
Microbial biogeography of wine grapes is conditioned by cultivar, vintage, and climate	2014	Proceedings of the National Academy of Sciences of the United States of America	274	2-s2.0-84891933123
Herbivores and nutrients control grassland plant diversity via light limitation	2014	Nature	264	2-s2.0-84899493708
SDMtoolbox: A python-based GIS toolkit for landscape genetic, biogeographic and species distribution model analyses	2014	Methods in Ecology and Evolution	258	2-s2.0-84904327892
A global assessment of precipitation chemistry and deposition of sulfur, nitrogen, sea salt, base cations, organic acids, acidity and pH, and phosphorus	2014	Atmospheric Environment	253	2-s2.0-84901847611
Hydrogen evolution from a copper(I) oxide photocathode coated with an amorphous molybdenum sulphide catalyst	2014	Nature Communications	242	2-s2.0-84906568523
Oxidation products of biogenic emissions contribute to nucleation of atmospheric particles	2014	Science	233	2-s2.0-84900445390
A global meta-analysis of the relative extent of intraspecific trait variation in plant communities	2015	Ecology Letters	217	2-s2.0-84946497490
The political origins of health inequity: Prospects for change	2014	The Lancet	217	2-s2.0-84893823818
Land-use intensity and the effects of organic farming on biodiversity: A hierarchical meta-analysis	2014	Journal of Applied Ecology	211	2-s2.0-84900535081
Salinisation of rivers: An urgent ecological issue	2013	Environmental Pollution	211	2-s2.0-84870229937
Biorefineries for the production of top building block chemicals and their derivatives	2015	Metabolic Engineering	198	2-s2.0-84911866672
The drivers of tropical speciation	2014	Nature	195	2-s2.0-84923809316

The analyses show that the update successfully removes articles from the field of biology, ecology and medicine. The co-occurrence of keywords shows a better focus on the targets of the SDG, even though some targets regarding policies, strategies and awareness raising seem to be missing or are inadequately covered. The second update could be used to add keywords on these targets to the search query.

Update 2) A second update was made to enhance the coverage of the different targets.

This second update was used to add keywords to the search query to enhance the coverage of the SDG's targets. The targets have been listed in the table below. Keywords have been identified that were not part of the search query yet but which could be essential to retrieve hits on the specific targets.

Target	Keywords to be added
Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries	(climate AND {disaster risk reduction})
Integrate climate change measures into national policies, strategies and planning	
Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning	{environmental education} – change education {sustainable development education} {environmental awareness} – change awareness {climate governance} (climate AND framework)
Implement the commitment undertaken by developed-country parties to the United Nations Framework Convention on Climate Change to a goal of mobilizing jointly \$100 billion annually by 2020 from all sources to address the needs of developing countries in the context of meaningful mitigation actions and transparency on implementation and fully operationalize the Green Climate Fund through its capitalization as soon as possible	{Green Climate Fund} {climate finance} {climate change financing} {climate investment} {climate capitalism}
Promote mechanisms for raising capacity for effective climate change-related planning and management in least developed countries and small island developing States , including focusing on women, youth and local and marginalized communities	(climate AND {small island developing states}) {climate justice} {climate equity}

In addition, duplicates (climate AND adaption) and (climate AND mitigation) were deleted. These query terms led to hits on several irrelevant articles, for example on adaptation in a school climate. Articles that are closely related to the theme of the SDG are retrieved by the query terms {climate adaptation} and {climate mitigation}.

Updated search query (v2): 157,241 document results

TITLE-ABS-KEY (({climate action} OR {climate adaptation} OR {climate change} OR {climate capitalism} OR ipcc OR {climate effect} OR {climate equity} OR {climate feedback} OR {climate finance} OR {climate change financing} OR {climate forcing} OR {climate governance} OR {climate impact} OR {climate investment} OR {climate justice} OR {climate mitigation} OR {climate model} OR {climate models} OR {climate modeling} OR {climate modelling} OR {climate policy} OR {climate policies} OR {climate risk} OR {climate risks} OR {climate services} OR {climate

service} OR {climate prediction} OR {climate predictions} OR {climate signal} OR {climate signals} OR {climate tipping point} OR {climate variation} OR {climate variations} OR ecoclimatology OR eco-climatology OR {Green Climate Fund} OR {regional climate} OR {regional climates} OR {urban climate} OR {urban climates} OR (climate AND ({adaptive management} OR awareness OR bioeconomy OR carbon OR {decision-making} OR {disaster risk reduction} OR {environmental education} OR {sustainable development education} OR {energy conservation} OR emission* OR extreme OR {food chain} OR {food chains} OR framework OR hazard* OR island* OR {land use} OR megacit* OR consumption OR production OR {small island developing states} OR anthropocene OR atmospher* OR {clean development mechanism} OR {glacier retreat} OR warming OR greenhouse OR {ice-ocean interaction} OR {ice-ocean interactions} OR {nitrogen cycle} OR {nitrogen cycles} OR {ocean acidification} OR {radiative forcing} OR {sea ice} OR {sea level} OR {sea levels} OR {thermal expansion} OR unfccc OR ozone))) AND NOT ({drug} OR {geomorphology})) AND PUBYEAR < 2018 AND PUBYEAR > 2012

The updated search returned 157,241 document results, compared to 158.963 document results for the original search.

The top 20 most cited articles were reviewed. Three of the 20 articles do not closely reflect the theme of the SDG. These articles covered topics related to land use, ecosystem services and ecology, but without clear relationships with climate change. Hits were found on the keywords 'land use' and 'climate change'.

Top 20 most cited articles listed in Scopus from search string version 2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Climate change 2013 the physical science basis: Working Group I contribution to the fifth assessment report of the intergovernmental panel on climate change	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	3502	2-s2.0-84926664150
High-resolution global maps of 21st-century forest cover change	2013	Science	2832	2-s2.0-84887724460
Updated high-resolution grids of monthly climatic observations - the CRU TS3.10 Dataset	2014	International Journal of Climatology	2469	2-s2.0-84896707401
Bounding the role of black carbon in the climate system: A scientific assessment	2013	Journal of Geophysical Research Atmospheres	1996	2-s2.0-84875359259
Planetary boundaries: Guiding human development on a changing planet	2015	Science	1624	2-s2.0-84922433390
Köppen's climate classification map for Brazil	2013	Meteorologische Zeitschrift	1528	2-s2.0-84897100402
High secondary aerosol contribution to particulate pollution during haze events in China	2015	Nature	1424	2-s2.0-84908317345
Increasing drought under global warming in observations and models	2013	Nature Climate Change	1274	2-s2.0-84871840877

Anthropogenic and natural radiative forcing	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1265	2-s2.0-84928644022
Changes in the global value of ecosystem services	2014	Global Environmental Change	1237	2-s2.0-84901406973
Long-term climate change: Projections, commitments and irreversibility	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	1199	2-s2.0-84928600768
Porous materials with optimal adsorption thermodynamics and kinetics for CO ₂ separation	2013	Nature	1126	2-s2.0-84874745873
Photocatalytic reduction of CO ₂ on TiO ₂ and other semiconductors	2013	Angewandte Chemie - International Edition	1072	2-s2.0-84880131318
The JRA-55 reanalysis: General specifications and basic characteristics	2015	Journal of the Meteorological Society of Japan	955	2-s2.0-84920667419
Observations: Atmosphere and surface	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	924	2-s2.0-84928600517
Carbon and other biogeochemical cycles	2013	Climate Change 2013 the Physical Science Basis: Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change	916	2-s2.0-84928642948
Status and ecological effects of the world's largest carnivores	2014	Science	912	2-s2.0-84892185014
The NCEP climate forecast system version 2	2014	Journal of Climate	900	2-s2.0-84896093136

Carbon capture and storage update	2014	Energy and Environmental Science	869	2-s2.0-84890526899
Impacts of ocean acidification on marine organisms: Quantifying sensitivities and interaction with warming	2013	Global Change Biology	826	2-s2.0-84877016612

In addition, a randomised sample of 20 articles was reviewed. Two of the 20 articles do not closely reflect the theme of this SDG. These two articles cover topics related to ecology and to historic human dispersal. Hits were found on the keywords 'climate' and 'climate conditions'.

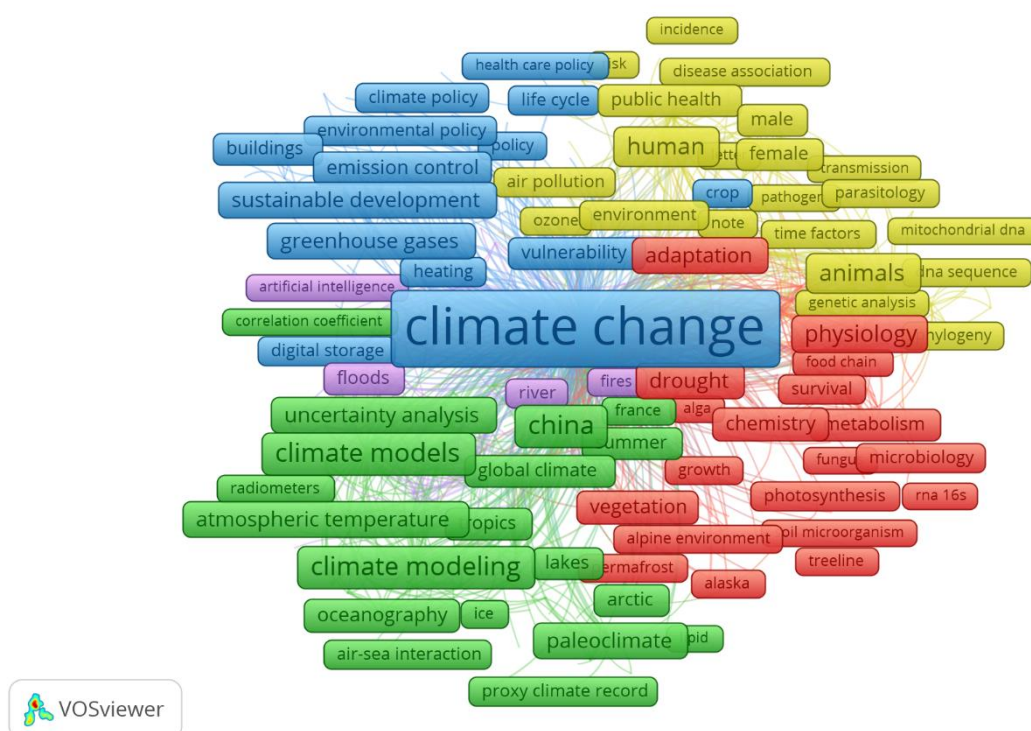
Randomised sample of 20 articles listed in Scopus from search string version 2:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
Drought over East Asia: A review	2015	Journal of Climate	83	2-s2.0-84949844533
Temperature-related mortality in 17 large Chinese cities: How heat and cold affect mortality in China	2014	Environmental Research	62	2-s2.0-84907326917
Influence of global climate change on chemical fate and bioaccumulation: The role of multimedia models	2013	Environmental Toxicology and Chemistry	54	2-s2.0-84871296940
Orbital-scale climate variability in Arabia as a potential motor for human dispersals	2015	Quaternary International	40	2-s2.0-84941792533
Interannual climate variability over the tropical pacific ocean induced by the indian ocean dipole through the Indonesian Throughflow	2013	Journal of Climate	34	2-s2.0-84877673786
Inter-comparison of high-resolution gridded climate data sets and their implication on hydrological model simulation over the Athabasca Watershed, Canada	2014	Hydrological Processes	31	2-s2.0-84903707174
Convergence of leaf-out towards minimum risk of freezing damage in temperate trees	2016	Functional Ecology	21	2-s2.0-84954405676
Simulation and economic analysis of indirect coal-to-liquid technology coupling carbon capture and storage	2013	Industrial and Engineering Chemistry Research	18	2-s2.0-84880916108
The effect of timing of growing season drought on flowering of a dominant C4 grass	2016	Oecologia	16	2-s2.0-84958767408
Multi-method attribution analysis of extreme precipitation in Boulder, Colorado	2016	Environmental Research Letters	14	2-s2.0-85008158217
Effects of temperature forcing provenance and extrapolation on the performance of an empirical glacier-melt model	2014	Arctic, Antarctic, and Alpine Research	12	2-s2.0-84902130914
Potential of <i>Trichogramma achaeae</i> (Hymenoptera: Trichogrammatidae) in Biological Control of <i>Tuta absoluta</i> (Lepidoptera: Gelechiidae) in Azorean Greenhouse Tomato Crops	2017	Journal of Economic Entomology	6	2-s2.0-85031819522
A diagram for evaluating multiple aspects of model performance in simulating vector fields	2016	Geoscientific Model Development	6	2-s2.0-85003781845

Modifications in vegetation cover and surface albedo during rapid urbanization: A case study from South China	2014	Environmental Earth Sciences	4	2-s2.0-84892879123
Distributions of "bomb 14C", biogeochemistry and elemental concentration in Hani mire peat profiles, NE China: Implications of environmental change	2017	Quaternary International	3	2-s2.0-85021058883
Modeling the influence of various water stressors on regional water supply infrastructures and their embodied energy	2016	Environmental Research Letters	3	2-s2.0-84977504111
Is Climate Change Shifting the Poleward Limit of Mangroves?	2017	Estuaries and Coasts	1	2-s2.0-85011282379
JPSS-1 VIIRS reflective solar band on-orbit calibration performance impacts due to SWIR nonlinearity artifacts	2016	Proceedings of SPIE - The International Society for Optical Engineering	1	2-s2.0-85010756817
Conservation research in the cloud forest of central America with lessons from Maderas Volcano, Ometepe, Nicaragua: A first-person narrative about very tough fieldwork, unfinished data, and climate justice while running out of time	2015	Central American Biodiversity: Conservation, Ecology, and a Sustainable Future	0	2-s2.0-84955379671
The investigation of solar energy conversion and storage processes in biomass	2013	8th International Conference on Electrical and Control Technologies, ECT 2013	0	2-s2.0-84941695513

Graphical summary of co-occurrence of terms from 2000 randomly selected results:



The new search results were compared to the search results retrieved by search query (v1) to check if the updated search was primarily removing irrelevant results.

As the update reduced the total number of document results, an additional analysis has been conducted to see whether mainly irrelevant document results have been removed.

The original and updated searches were combined using AND NOT function in Scopus. This returned 1,722 document results – i.e. the publications that were excluded from the first search in the second search.

The top 20 cited articles were reviewed. Nineteen of the 20 articles are rightfully excluded from the updated document results. These articles covered topics regarding educational climate, safe working climate, adaptation or mitigation in ecological studies and healthcare.

Top 20 cited articles included in the search string v1 results but not in the updated search results:

*red text shows articles which do not closely match the theme of the SDG

Title	Year	Source title	Cited by	EID
A Review of School Climate Research	2013	Review of Educational Research	408	2-s2.0-84881009463
Environmental heterogeneity as a universal driver of species richness across taxa, biomes and spatial scales	2014	Ecology Letters	389	2-s2.0-84901694925
Hand, foot, and mouth disease in China, 2008-12: An epidemiological study	2014	The Lancet Infectious Diseases	383	2-s2.0-84896492664
A global analysis of the impacts of urbanization on bird and plant diversity reveals key anthropogenic drivers	2014	Proceedings of the Royal Society B: Biological Sciences	378	2-s2.0-84893711703
Leaf venation: Structure, function, development, evolution, ecology and applications in the past, present and future	2013	New Phytologist	255	2-s2.0-84877602457
The rise of women: The growing gender gap in education and what it means for American schools	2013	The Rise of Women: The Growing Gender Gap in Education and what it Means for American Schools	234	2-s2.0-84917665723
The NAACP's legal strategy against segregated education, 1925-1950	2014	The NAACP's Legal Strategy against Segregated Education, 1925-1950	195	2-s2.0-85028412514
Testing for associations between loci and environmental gradients using latent factor mixed models	2013	Molecular Biology and Evolution	193	2-s2.0-84879335655
Impacts of invasive alien marine species on ecosystem services and biodiversity: A pan-European review	2014	Aquatic Invasions	182	2-s2.0-84912046427
Agriculture facilitated permanent human occupation of the Tibetan Plateau after 3600 B.P	2015	Science	159	2-s2.0-84922480781
Culture of Disengagement in Engineering Education?	2014	Science Technology and Human Values	145	2-s2.0-85032069153
Population genomics reveal recent speciation and rapid evolutionary adaptation in polar bears	2014	Cell	131	2-s2.0-84900301308

Environmental flows for natural, hybrid, and novel riverine ecosystems in a changing world	2014	Frontiers in Ecology and the Environment	127	2-s2.0-84907549868
Heat-related mortality and adaptation to heat in the United States	2014	Environmental Health Perspectives	125	2-s2.0-84905216322
Genomic insights into salt adaptation in a desert poplar	2013	Nature Communications	113	2-s2.0-84888247172
In situ simulation: Detection of safety threats and teamwork training in a high risk emergency department	2013	BMJ Quality and Safety	107	2-s2.0-84878343656
Academic Research in the 21st Century: Maintaining Scientific Integrity in a Climate of Perverse Incentives and Hypercompetition	2017	Environmental Engineering Science	99	2-s2.0-85018260001
Optimization of micro-grid system using MOPSO	2014	Renewable Energy	99	2-s2.0-84902328636
Adaptation and possible ancient interspecies introgression in pigs identified by whole-genome sequencing	2015	Nature Genetics	95	2-s2.0-84924026228
A perfect storm: How tumor biology, genomics, and health care delivery patterns collide to create a racial survival disparity in breast cancer and proposed interventions for change	2015	CA Cancer Journal for Clinicians	95	2-s2.0-84929128699

The second update has successfully added topics to the document results. This was indicated by findings of the randomised sample, e.g. the article on climate justice. However, this is not clearly reflected in the co-occurrence of terms. This could be explained by an overly focus of scientific research on climate modelling and related topics. The update also excluded articles that are not related to the theme of the SDG. The current search query (v2) appears to be an improvement compared to the original search query.